

Neck Pain

By **Peter J. Moley**, MD, Hospital for Special Surgery

Reviewed/Revised Nov 2024 | Modified Aug 2025

Along with [low back pain](#), neck pain is a very common reason for health care visits. The pain usually results from problems with the musculoskeletal system—the spine, including the bones of the spine (backbones, or vertebrae) and the muscles and ligaments that support it. Some disorders cause only neck pain. Other disorders can cause neck and low back pain. Occasionally, neck pain results from a disorder, such as [meningitis](#), that does not involve the musculoskeletal system.

The neck's flexibility makes it susceptible to wear and tear and to injuries that overstretch it, such as whiplash. Also, the neck has the critical job of holding up the head. Poor posture makes that job more difficult. Neck pain, like back pain, is common and becomes more common as people age. For pain located in the front of the neck, see [Sore Throat](#).

The part of the spine that is in the neck is called the cervical spine. It consists of 7 bones (vertebrae), which are separated by discs made of jelly-like material and surrounded by cartilage. The cervical spine contains a canal that houses the [spinal cord](#). Along the length of the spinal cord, spinal nerves emerge through spaces between the vertebrae to connect with nerves throughout the body. The part of the spinal nerve nearest the spinal cord is the spinal nerve root. Muscles and ligaments in the neck support the spine.

Neck pain can involve damage to bones, muscles, discs, or ligaments, but pain can also be caused by damage to nerves or the spinal cord. A spinal nerve root can be compressed when the spine is injured, resulting in pain and sometimes weakness, numbness, and tingling in an arm. [Compression of the spinal cord](#) can cause numbness and weakness of both arms and both legs and rarely loss of bladder control ([urinary incontinence](#)) and bowel control ([fecal incontinence](#)).



Causes of Neck Pain

Most of the [disorders that can cause low back pain](#) can also cause neck pain, and most involve the spine, the tissues that support it, or both.

Common causes

Common causes of neck pain include

- Muscle spasms
- Injuries to muscles and ligaments
- Arthritis (usually [osteoarthritis](#))
- [Cervical spondylosis](#)
- [Cervical spinal stenosis](#)
- [A ruptured or herniated disc](#)
- [Fibromyalgia](#)

Spasms of the neck muscles are common and may occur on their own or after an injury, even a minor injury.

Injuries may occur during routine activities (for example, lifting, exercising, moving in an unexpected way) or result from trauma such as a fall or car crash. Often no specific injured structures are identified with imaging tests, but doctors presume that some muscles and/or ligaments have been affected.

In [cervical spondylosis](#), the vertebrae in the neck, joints of the neck (facet joints), and the discs between them degenerate, usually because of [osteoarthritis](#). As a result, the

nerves that emerge through the vertebrae may be pinched. Sometimes the spinal canal is narrowed (cervical spinal stenosis), and the spinal cord is compressed.

[Cervical spinal stenosis](#) describes the condition in which the space inside the spine canal becomes narrowed. In the cervical spine this results in compressing the thecal sac (the outer covering of the spinal cord), often the spinal cord, and/or its nerve roots. The usual cause is a combination of [osteoarthritis](#), disc degeneration, and thickening of the soft tissues in the neck.

[Herniated discs](#) can cause neck pain. The discs between each of the vertebrae have an outer layer of cartilage (a tough, fiber-like tissue) and a soft, jelly-like interior. If a disc is repeatedly squeezed by the vertebrae above and below it, the outer layer may tear (rupture), causing pain. The interior of the disc can bulge out through the tear (herniate). The bulging disc can push on or even damage the spinal nerve root next to it. Rarely, the disc compresses the spinal cord.

[Fibromyalgia](#) is a cause of pain, sometimes including neck pain. This disorder causes chronic widespread pain in muscles and other soft tissues in areas besides the neck.

Less common causes

Less common causes of neck pain that are serious include

- A tear in the lining of a neck artery (dissection)
- [Meningitis](#)
- A [spinal tumor](#) or infection
- A [heart attack](#) or [angina](#) (chest pain due to an inadequate blood supply to the heart muscle)

[Spasmodic torticollis](#) is also a less common cause but is not as serious as some causes. It is a severe type of spasm of a specific neck muscle that causes the head to tilt and rotate into an abnormal position. Sometimes the spasms are rhythmic, causing the head to jerk. The cause may be unknown or may be due to certain medications or hereditary disorders.

Evaluation of Neck Pain

The following information can help people with neck pain decide whether a doctor's evaluation is needed and help them know what to expect during the evaluation. In the evaluation, doctors first try to identify serious disorders.

Warning signs

In people with neck pain, certain signs are cause for concern. They include

- Loss of strength or sensation in the arms and legs—possibly a symptom of nerve damage
- Fever
- Night sweats
- Headache
- Lethargy or confusion
- Chest discomfort
- Sudden sweating or difficulty breathing
- Pain that is triggered by exertion or worsens during exertion

When to see a doctor

People with warning signs or difficulty or pain when swallowing should see a doctor immediately.

If people without warning signs have severe pain (particularly if it is not relieved by [acetaminophen](#) or a [nonsteroidal anti-inflammatory drug](#) [NSAID]), they should see a doctor within a day or so.

Other people can wait a few days or call their doctor to discuss how soon they need to be seen.

What the doctor does

Doctors first ask questions about the person's symptoms and medical history. Doctors then do a physical examination. What they find during the history and examination often suggests a cause of the neck pain and the tests that need to be done (see table [Some Causes and Features of Neck Pain](#)).

The physical examination focuses on the spine and nervous system ([neurologic examination](#)) to look for signs of nerve root or [spinal cord compression](#). Signs of nerve root compression include muscle weakness, abnormal reflexes (tested by tapping the tendons around the elbow and wrist), decreased sensation in parts of the body other than the head, [inability to urinate](#), and [incontinence of urine](#) or [incontinence of stool](#). Doctors may ask the person to move the neck in certain ways.

With information about the pain, the person's medical history, and results of a physical examination, doctors may be able to determine the most likely causes:

- Loss of strength or sensation may indicate damage to the spine nerves exiting from the cervical spine.
- Pain on the front or on one side of the neck usually is not caused by a problem with the spinal cord.
- Neck pain that radiates down an arm is usually caused by cervical disc herniation, cervical spinal stenosis, or spondylosis with compression of the spinal nerve root.
- Pain in the chest that travels up into the neck or into the arm may be caused by a heart disorder such as [angina](#) or a [heart attack](#).
- Pain that is constant, severe, progressively worse, and unrelieved by rest, particularly if it keeps the person awake at night and is accompanied by sweating, may indicate cancer or an infection.



Some Causes and Features of Neck Pain

Cause	Common Features*	Tests
More common but less serious causes		
Sprains and strains	Pain that • Usually occurs off to one side of the spine • Worsens with movement and lessens with rest	A doctor's examination
Muscle spasms	Constant pain and stiffness, causing difficulty or pain when the head is turned one way or the other (sometimes both ways) No symptoms of nervous system malfunction (neurologic symptoms)	A doctor's examination
<u>Osteoarthritis</u> (without compression of the spinal nerve root)	Pain that • Is sometimes constant • Worsens with motion	A doctor's examination
 MSD © 2026 Merck & Co., Inc., Rahway, NJ, USA and its affiliates. All rights reserved.	Often in people with arthritis in the joints of fingers, hips, and/or knees	Sometimes neck x-rays
<u>Cervical spondylosis</u> (with compression of the spinal nerve root)	Pain that • Often extends down the arm,	MRI of the neck